Semi-supervised Coupled Dictionary Learning for Cross-modal Retrieval in Internet Images and Texts

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ABSTRACT

Nowadays massive amount of images and texts has been emerging on the Internet, arousing the demand of effective cross-modal retrieval. To eliminate the heterogeneity between the modalities of images and texts, the existing subspace learning methods try to learn a common latent subspace under which cross-modal matching can be performed. However, these methods usually require fully paired samples (images with corresponding texts) and also ignore the class label information along with the paired samples. Indeed, the class label information can reduce the semantic gap between different modalities and explicitly guide the subspace learning procedure. In addition, the large quantities of unpaired samples (images or texts) may provide useful side information to enrich the representations from learned subspace. Thus, in this paper we propose a novel model for cross-modal retrieval problem. It consists of 1) a semi-supervised coupled dictionary learning step to generate homogeneously sparse representations for different modalities based on both paired and unpaired samples; 2) a coupled feature mapping step to project the sparse representations of different modalities into a common subspace defined by class label information to perform cross-modal matching. Experiments on a large scale web image dataset MIRFlickr-1M with both fully paired and unpaired settings show the effectiveness of the proposed model on the cross-modal retrieval task.

Categories and Subject Descriptors

H.3.3 [**Information Storage and Retrieval**]: Information Search and Retrieval

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General Terms

Algorithms, Experimentation, Performance

Keywords

Cross-modal Retrieval, Semi-supervised learning, Coupled Dictionary Learning

1. INTRODUCTION

During the last decade, there has been a tremendous volume of multimedia data emerging on the Internet. Usually, the data come from different input channels and are usually paired with one-to-one correspondences. For examples, images are associated with texts, videos contain visual and audio signals. Recently, jointly modeling the Internet images and associated texts has gained much attentions in the multimedia research field [7, 9, 10], resulting in a variety of cross-modal retrieval scenarios: 1) "*Text2Img*": given a text query (document or several words), finding the most related image; 2) "*Img2Text*": given an image query, finding the words or phrases that best describe the image. Indeed, there are several challenges existing in the cross-modal retrieval problem: 1) the images and texts have different representations. For example, images are usually represented using real-valued and dense feature descriptors, whereas texts are represented as discrete sparse word count vectors; 2) the different modalities of visual and textual features cannot be matched directly with each other since they have distinct statistical properties; 3) the high level semantic description of texts and the extracted low level visual descriptors lead to semantic gap between the modalities of images and texts.

To eliminate the diversities between the different modalities, a number of recent approaches that are are concentrated on latent subspace learning have been proposed, such as Canonical Correlation Analysis (CCA) [3] and its extensions [1, 2, 6]. These methods are designed to learn a common subspace, in which the correlations in paired samples can also be well preserved and the projected features of different modalities can be measured directly. However, these methods have two limitations: 1) they do not utilize the class label information (e.g., categories or attributes of im-

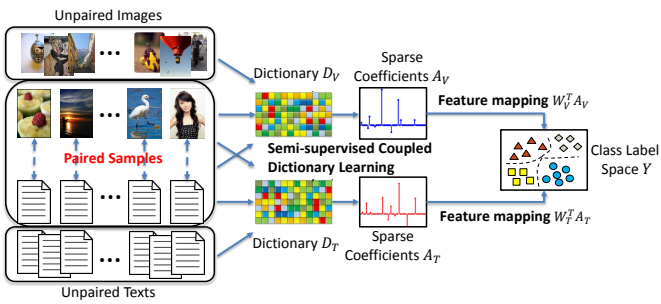


Figure 1: Flowchart of the proposed model.

ages) along with the paired samples; 2) they require fully paired training samples, where a paired sample should consist of an image and its corresponding text. They cannot handle the more common data on the web that are weakly paired or unpaired, such as images with noisy texts or without texts. Indeed, the class label information would be very helpful to explicitly guide the subspace learning and reduce the semantic gap between the modalities of images and texts. And the unpaired samples can provide useful side information and enrich the representations of learned subspace.

In this work, to tackle the limitations of these methods, we propose a novel model for cross-modal retrieval problem. The proposed model, as shown in Fig. 1, consists of two steps: *semi-supervised coupled dictionary learning* and *coupled feature mapping*. In the first step, to deal with paired and unpaired samples simultaneously, we develop a semi-supervised coupled dictionary learning algorithm that transforms the original features of different modalities into homogeneously sparse representations. In the second step, with the class label information, we derive an efficient feature mapping scheme that projects the learned sparse representations of the paired samples into a discriminant subspace defined by class information.

The novelty and advantages of our work include: 1) a practical solution that can efficiently handle paired and unpaired data, as well as the valuable class information; 2) a novel model with different methodology based on semi-supervised coupled dictionary learning and feature mapping schemes, comparing with the most recent works [5] (facilitating blob based image features) and [7] (using deep learning architecture). Experiments on a large scale web image dataset with both fully paired and unpaired settings show the effectiveness of the proposed model.

The rest of the paper is organized as follows. In Section 2, we describe the details of the proposed model. Section 3 reports the experimental results and the conclusion is summarized in Section 4.

2. THE PROPOSED MODEL

Let $\mathbf{Z} = \{\mathbf{Z}^p \cup \mathbf{Z}^u\}$ denote a collection of training samples with features from two different modalities. $\mathbf{Z}^p = \{(\mathbf{v}_i, \mathbf{t}_i)\}|_{i=1}^{N^p}$ contains the paired samples, such as the images and their associated texts, where a paired sample $(\mathbf{v}_i, \mathbf{t}_i)$ consists of d_1 dimensional visual feature $\mathbf{v}_i$ and d_2 dimensional text feature $\mathbf{t}_i$. $\mathbf{V}^p = [\mathbf{v}_1, ..., \mathbf{v}_{N^p}] \in \mathbb{R}^{d_1 \times N^p}$ and $\mathbf{T}^p = [\mathbf{t}_1, ..., \mathbf{t}_{N^p}] \in \mathbb{R}^{d_2 \times N^p}$ are feature matrices of paired images and texts, respectively. $\mathbf{Z}^u = \{(\mathbf{v}_j,)|_{j=1}^{N_v^u} \cup (, \mathbf{t}_k)|_{k=1}^{N_t^u}\}$ is the subset of N^u unpaired samples including N_v^u images without associated texts and N_t^u texts without corresponding images. Similarly, $\mathbf{V}^u \in \mathbb{R}^{d_1 \times N_v^u}$ and $\mathbf{T}^u \in \mathbb{R}^{d_2 \times N_t^u}$ are feature matrix of unpaired images and texts, respectively. Specifically, for the top m paired samples in $\mathbf{Z}^p$ $(m \leq N^p)$, besides the one-to-one correspondences of visual and text features, they also have class label information $\{\mathbf{y}_i\}_{i=1}^t$. Here $\mathbf{y}_i \in \{0,1\}^{c \times 1}$ is a c dimensional binary class indicator vector.

For the cross-modal retrieval problems, our primary goal is to learn a representative latent subspace that can get rid of the heterogeneity between different modalities and incorporate the relationships across modalities, by utilizing both paired and unpaired training samples. In addition, an explicit mapping function is also required to ensure the learned space to be discriminant based on the class information of the paired training samples.

2.1 Semi-supervised Coupled Dictionary Learning

To handle both paired and unpaired samples of two different modalities, we employ sparse representation from dictionary learning for each modality since it has been shown to be very effective in data representation and reconstruction problems. Specifically, we introduce a semi-supervised coupled dictionary learning method that learns two two dictionaries $\mathbf{D}_V \in \mathbb{R}^{d_1 \times k_1}$ and $\mathbf{D}_T \in \mathbb{R}^{d_2 \times k_2}$ of two modalities, and the paired samples are used to carry the relationship between different modalities while the unpaired samples are introduced to exploit the marginal distribution for obtaining robust sparse representations. The proposed method can be formulated to minimize the following objective function

$$E(\mathbf{D}_V, \mathbf{D}_T, \mathbf{A}_V, \mathbf{A}_T) = E^u(\mathbf{D}_V, \mathbf{A}_V^u) + E^u(\mathbf{D}_T, \mathbf{A}_T^u) + E^p(\mathbf{D}_V, \mathbf{D}_T, \mathbf{A}_V^p, \mathbf{A}_T^p), \quad (1)$$

where $\{\mathbf{A}_V^p, \mathbf{A}_T^p\}$ and $\{\mathbf{A}_V^u, \mathbf{A}_T^u\}$ are the sparse coefficients for paired and unpaired samples of two modalities, respectively. Specifically, to guarantee that the sparse representations of the two modalities well reconstruct the unpaired samples, in Eq. 1 we have

$$E^u(\mathbf{D}_V, \mathbf{A}_V^u) = \|\mathbf{V}^u - \mathbf{D}_V \mathbf{A}_V^u\|_F^2 + \sigma_V^u \|\mathbf{A}_V^u\|_1, \quad (2)$$

$$E^u(\mathbf{D}_T, \mathbf{A}_T^u) = \|\mathbf{T}^u - \mathbf{D}_T \mathbf{A}_T^u\|_F^2 + \sigma_T^u \|\mathbf{A}_T^u\|_1, \quad (3)$$

which are standard form of sparse coding. And the useful information in unpaired samples can be reflected in the dictionaries of the two modalities. Furthermore, to ensure the dictionaries of two modalities to be coupled, we impose a function $f(\mathbf{A}_V^p, \mathbf{A}_T^p)$ to relate the sparse representations of two modalities of the paired samples. Then for the paired samples, in Eq. 1, we have

$$E^p(\mathbf{D}_V, \mathbf{D}_T, \mathbf{A}_V^p, \mathbf{A}_T^p) = \|\mathbf{V}^p - \mathbf{D}_V \mathbf{A}_V^p\|_F^2 + \|\mathbf{T}^p - \mathbf{D}_T \mathbf{A}_T^p\|_F^2 + \sigma^p(\|\mathbf{A}_V^p\|_1 + \|\mathbf{A}_T^p\|_1) + f(\mathbf{A}_V^p, \mathbf{A}_T^p), \quad (4)$$
$$s.t. \quad \|\mathbf{d}_{v,i}\|_2 \leq 1, \|\mathbf{d}_{t,i}\|_2 \leq 1, \forall i.$$

Inspired by [4], we introduce a k_c dimensional common feature space $\mathcal{P}$ for $f(\mathbf{A}_V^p, \mathbf{A}_T^p)$ so that each pair of samples from $\mathbf{A}_V^p$ and $\mathbf{A}_T^p$ can be transformed to the points of $\mathbf{P}_T$ and $\mathbf{P}_V$ in space $\mathcal{P}$ mutually. Here we restrict $k_c = k_1 = k_2$ so that different modalities of data are comparable in $\mathcal{P}$.

The formulation of $f(\mathbf{A}_V^p, \mathbf{A}_T^p)$ is derived as

$$f(\mathbf{A}_V^p, \mathbf{A}_T^p) = \gamma(\|\mathbf{A}_V^p - \mathbf{U}_V^{-1}\mathbf{P}_T\|_F^2 + \|\mathbf{A}_T^p - \mathbf{U}_T^{-1}\mathbf{P}_V\|_F^2) + \mu(\|\mathbf{U}_V^{-1}\|_F^2 + \|\mathbf{U}_T^{-1}\|_F^2), \quad (5)$$

where $\mathbf{U}_V \in \mathbb{R}^{k_c \times k_1}$ and $\mathbf{U}_T \in \mathbb{R}^{k_c \times k_2}$ are the transform matrices for two modalities. The regularized constraints on $\mathbf{U}_V$ and $\mathbf{U}_T$ ensure numerical stability and avoid overfitting. It can be learned that the constraints in Eq. 5 shows the ability of recovering the sparse representations in one modality using data from the other, hence the relationship across different modalities can be efficiently incorporated in the sparse representations.

2.2 Coupled Feature Mapping

Let $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, ..., \mathbf{y}_n] \in \mathbb{R}^{c \times m}$ be the class label matrix of the paired samples $\{(\mathbf{v}_i, \mathbf{t}_i)\}|_{i=1}^m$. The coupled feature mapping aims to learn two projection matrices $\mathbf{W}_V \in \mathbb{R}^{k_1 \times c}$ and $\mathbf{W}_T \in \mathbb{R}^{k_2 \times c}$, which map the sparse representations $\mathbf{A}_{v_i}^p|_{i=1}^m$ and $\mathbf{A}_{t_i}^p|_{i=1}^m$ of the two modalities for the paired samples into the common subspace defined by class labels.

In order to minimize the errors of projecting the sparse representations of each modality to the label space, besides the basic ridge regression form, we also employ the coupled feature selection scheme proposed in [8], to select relevant and discriminative features from the sparse representations and to enhance the relevance across different modalities. The objective function of the coupled feature mapping procedure can be derived as

$$\min_{\mathbf{W}_V, \mathbf{W}_T} \frac{1}{2}(\|\mathbf{W}_V^\top \mathbf{A}_V^p - \mathbf{Y}\|_F^2 + \|\mathbf{W}_T^\top \mathbf{A}_T^p - \mathbf{Y}\|_F^2) \quad (6)$$
$$+ \lambda_1(\|\mathbf{W}_V\|_{21} + \|\mathbf{W}_T\|_{21}) + \lambda_2\|[\mathbf{W}_V^\top \mathbf{A}_V^p \mathbf{W}_T^\top \mathbf{A}_T^p]\|_*,$$

where the L_{21} norm $\|\cdot\|_{21}$ is used to select features from coupled feature space jointly, the trace norm $\|\cdot\|_*$ encodes the correlations of different modalities with a low-rank constraint. Note that Eq. 6 differs from [8] in that coupled feature selection is performed on sparse representations $\mathbf{A}_V^p$ and $\mathbf{A}_T^p$ to leverage the merit from coupled dictionary learning, rather than on the original features in [8].

2.3 Solution

The objective function in Eq. 1 is convex with each of the variables of dictionaries $\{\mathbf{D}_V, \mathbf{D}_T\}$, sparse representations $\{\mathbf{A}_V, \mathbf{A}_T\}$ and transform matrices $\{\mathbf{U}_V, \mathbf{U}_T\}$ when fixing the other variables. They can be iteratively updated in an alternating manner. Similar as in [4], We utilize the sparse modeling software SPAMS[1] to efficiently implement the optimization algorithm. In addition, we use the iterative algorithm developed in [8] to optimize Eq. 6.

In the testing phase, given a paired sample $(\hat{\mathbf{v}}, \hat{\mathbf{t}})$, we first generate the sparse representations $\hat{\mathbf{A}}_{\hat{v}}$ and $\hat{\mathbf{A}}_{\hat{t}}$ based on learned dictionaries $\mathbf{D}_V$ and $\mathbf{D}_T$, respectively. Then we can project $\hat{\mathbf{A}}_{\hat{v}}$ and $\hat{\mathbf{A}}_{\hat{t}}$ into the class label space through the learned projection matrices $\mathbf{W}_V$ and $\mathbf{W}_T$, respectively. To perform cross-modal retrieval, we take a sample of one modal data (i.e., an image) as the query to retrieval the other modality of it (i.e., the texts). Even if an unpaired sample with one missing modality is given, we can still reconstruct its missing modality using the similar pipeline.

[1]http://spams-devel.gforge.inria.fr/

3. EXPERIMENTS

We evaluate the effectiveness of the proposed model on a large scale web image dataset MIRFlickr-1M. This dataset consists of one million web images retrieved from Flickr website along with their user assigned tags. Among the 1 million images, a subset of 25K images have been manually annotated with 38 class labels such as object and scene categories, where each image may belong to several classes. The remaining subset of 975K images have no class labels. For the feature representation of image and text, we use the features extracted in [7]. Specifically, each image is represented by 3,857 dimensional features by concatenating SIFT, Gist and MPEG-7 descriptors. Each text input is represented using a vocabulary of the 2,000 most frequent tags in the 1 million collection, and the average number of tags associated with an image is 5. Since there are 128,501 images which do not have any tags, thus in our experiment the number of paired and unpaired samples are $N^p = 871,499$ and $N_v^u = 128,501$, respectively. For the evaluation, the standard metric of mean average precision (MAP) is used for the cross-modal retrieval: Img2Text and Text2Img tasks. Besides, we also evaluate our model on classification task for the paired samples with class labels and the results are reported using MAP of samples and top-50 predictions (Prec@50) for multi-label classification. Note that for all the metrics, larger numerical value indicates better performance. In the training stage, we empirically set the parameters σ, γ, μ and dictionary size to 0.01, 0.0001, 0.001, 350 for the semi-supervised coupled dictionary learning procedure; and the parameters λ_1 and λ_2 to 0.1 and 0.001 for the coupled feature mapping procedure, respectively. In testing phase, the cosine distance is adopt as similarity measure to select the best matches.

3.1 Results on Fully Paired Setting

We first evaluate the proposed model on fully paired setting. Specifically, we use the subset of the 25K paired samples with class labels and randomly allocate 10K, 5K, 10K images for training, validation and testing, respectively. And we report the averaged results over five random splits of the 25K paired samples. Three typical subspace learning methods of Kernel CCA (KCCA) [1], MultiView CCA (MCCA) [2] and LCFS [8] are compared with the proposed semi-supervised dictionary learning scheme. Then a baseline using ridge regression (RR), two recent works of LGCFL [5] and DBM [7] are compared with the proposed model. We implement the KCCA and LGCFL algorithm according to the instructions in the corresponding papers and refer to the code of the other methods from the authors' websites.

Task	Cross-modal retrieval		Classification	
Method	Img2Text	Text2Img	MAP	Prec@50
KCCA [1]	0.579	0.589	-	-
MCCA [2]	0.557	0.551	-	-
RR	-	-	0.432	0.712
LCFS [8]	0.533	0.546	0.449	0.762
LGCFL [5]	0.596	0.599	0.504	0.811
DBM [7]	0.600	0.607	0.526	0.791
Proposed	0.612	0.624	0.532	0.824

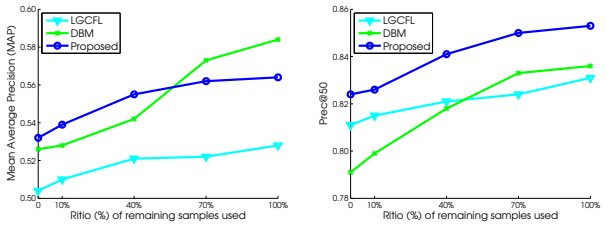
Table 1: Overall results on the subset of 25K paired samples under fully paired setting.

Table 1 shows the overall performance of these methods on cross-modal retrieval task and classification task. On the cross-modal task, the proposed model achieves higher MAP

scores than the subspace learning methods of KCCA, MCCA and LCFS, showing the effectiveness of the coupled dictionary learning scheme in the proposed model on learning representative subspace. In addition, the proposed model also outperforms the two recent works of LGCFL and DBM, indicating that the learned sparse representations in the proposed model is effective and reliable to get rid of the diversity between different modalities on the paired setting. On the classification task, the propose model performs much better than the baseline of RR that only uses image modality to learn model, showing that combining both image and text modalities improves the accuracy of classification task than using single image modality. Moreover, the proposed model achieves the best performance comparing with the other methods, indicating that with small number of paired training samples, the proposed model is able to enhance the relationship across different modalities by leveraging the useful class label information.

3.2 Results on Partially Paired Setting

We then evaluate the proposed model on partially paired setting, with the training samples containing both paired samples and unpaired samples. We consider the subset of 25K paired samples with class labels as the start point, and gradually add the remaining paired and unpaired samples for training until covering the entire 1 million samples. We still report the evaluation results on the 25K subset. The proposed model is compared with the LGCFL and DBM methods that can also handle partially paired setting. Specifically, for the LGCFL method we first use the model trained on the 25K subset to predict class labels for the remaining samples, and then conduct experiment on the entire dataset of 1 million samples with partially paired setting.



(a) MAP (b) Prec@50

Figure 2: Effects of using different ratio of remaining samples on classification task.

Fig. 2 shows the MAP and Prec@50 scores on classification task with different ratio of remaining samples added gradually, from 10% to 100%. We can observe that for all three methods further improvements can be made by using more additional data. In particular, for the proposed model, the performance consistently increases when more samples are added. The reason is that the semi-supervised learning manner can incorporate the unimodal data of unpaired samples in the coupled dictionary learning step, which enhances to obtain robust sparse representations for both modalities. The proposed model performs better than the LGCFL method and it is more robust to tackle heterogeneous feature representations of different modalities, whereas the LGCFL method is elaborately designed for blob based image features and requires class label information even for unpaired samples. Moreover, the proposed model obtains comparable result with the DBM method, which uses deep Boltzmann machine to learn joint space of image and text modalities. Compared with the sophisticated deep architecture in DBM, the proposed model is simpler and promising to cope with large scale data and to obtain competitive performance under the partially paired setting. Table 2 depicts the overall

Task	Cross-modal retrieval		Classification	
Method	Img2Text	Text2Img	MAP	Prec@50
LGCFL [5]	0.598	0.601	0.528	0.831
DBM [7]	0.602	0.611	0.584	0.836
Proposed	0.619	0.631	0.564	0.853
Proposed (CNN)	0.640	0.652	0.678	0.884

Table 2: Overall results on entire collection of 1 million samples under partially paired setting.

results of the three method on the entire collection of 1 million samples under partially paired setting. And we can obtain similar observation from Table 2 that the proposed model performs comparable with the two recent works and further benefits from using features extracted from the popular VGG Net of deep convolutional neural network.

4. CONCLUSION

In this paper, we propose a novel model solving the practical cross-modal retrieval problem. To handle both paired and unpaired samples simultaneously, we propose a semi-supervised coupled dictionary learning scheme to generate homogeneously sparse representations for different modalities. We then use coupled feature mapping scheme to project the sparse representations of different modalities into the class label space for cross-modal matching. Experiments on a large scale web image dataset with both fully paired and unpaired settings show the effectiveness of our proposal.

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5. REFERENCES

[1] L. Ballan, T. Uricchio, L. Seidenari, and A. Bimbo. A cross media model for automatic image annotation. In *ICMR*, 2014.

[2] Y. Gong, Q. Ke, M. Isard, and S. Lazebnik. A multi-view embedding space for modeling internet images, tags, and their semantics. *IJCV*, 106:210–233, 2014.

[3] D. R. Hardoon, S. R. Szedmak, and J. R. Shawe-taylor. Canonical correlation analysis: An overview with application to learning methods. *Neural Comput.*, 16(12):2639–2664, 2004.

[4] D. Huang and Y. Wang. Coupled dictionary and feature space learning with applications to cross-domain image synthesis and recognition. In *ICCV*, pages 2496–2503, 2013.

[5] C. Kang, S. Xiang, S. Liao, C. Xu, and C. Pan. Learning consistent feature representation for cross-modal multimedia retrieval. *TMM*, 17(3):370–381, 2015.

[6] N. Rasiwasia, J. Costa Pereira, E. Coviello, G. Doyle, G. Lanckriet, R. Levy, and N. Vasconcelos. A new approach to cross-modal multimedia retrieval. In *ACM MM*, 2010.

[7] N. Srivastava and R. Salakhutdinov. Multimodal learning with deep boltzmann machines. *JMLR*, 15:2949–2980, 2014.

[8] K. Wang, R. He, W. Wang, L. Wang, and T. Tan. Learning coupled feature spaces for cross-modal matching. In *ICCV*, pages 2088–2095, 2013.

[9] Y. Yang, Y. Yang, and H. T. Shen. Effective transfer tagging from image to video. *ACM Trans. Multimedia Comput. Commun. Appl.*, 9(2):1–20, 2013.

[10] Y. Yang, Z.-J. Zha, Y. Gao, X. Zhu, and T.-S. Chua. Exploiting web images for semantic video indexing via robust sample-specific loss. *TMM*, 17(2):246–256, 2015.